

Week 9 — Activity: Plate Cooling and Fourier Series

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Purpose

To understand how imposing a fixed lithospheric thickness modifies transient cooling and why Fourier series naturally arise in plate cooling models.

Concept

Plate models introduce a basal boundary condition at a fixed depth, representing the lithosphere–asthenosphere boundary.

Mathematical Form

The temperature field can be expressed as a Fourier series:

$$T(z, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi z}{L}\right) \exp\left(-\frac{n^2 \pi^2 \kappa t}{L^2}\right).$$

Tasks

1. Explain how the index n relates to thermal wavelength.
2. Why do short-wavelength modes decay faster than long-wavelength modes?
3. Compare thermal evolution predicted by plate cooling and half-space cooling.

Geological Interpretation

- Plate thickness limits lithospheric growth.
- Explains flattening of bathymetry and heat flow for old oceanic plates.

Connection

Relate Fourier modes to spectral filtering concepts used previously in gravity and magnetic field analysis.